Reg. No. | | | | | | | | | | | [17EE42]

FOURTH SEMESTER B. Tech EXAMINATION, JUNE – 2019

MICROPROCESSORS AND MICROCONTROLLERS

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions choosing at least TWO Questions from each part.
2. Missing data may be suitably assumed.

PART - A

1. a. Draw the block diagram of 8086 Microprocessor and explain working of Execution Unit and Bus Interface Unit. **10**

   b. Discuss the bi-functionality of AX, BX CX and DX registers. **04**

2. a. Discuss the following Instructions with example.

   i. ADD *dest,src* **06**

   ii. XLAT

   iii. LAHF

   b. Write an Assembly Language Program to find average of 'N' given 16-bit numbers. **08**

3. a. Explain the following Addressing mode of 8086 microprocessor.

   i. Direct Addressing Mode **06**

   ii. Indirect Addressing Mode

   iii. Register Indirect Mode

   b. Describe the action taken by 8086 microprocessor on interrupt request. **08**

4. a. Write an Assembly Language Program to generate sinewave **06**

   b. Interface a stepper motor and write an Assembly Language Program to rotate in clock-wise and anti-clock wise direction. **08**

PART - B

5. a. Discuss the function of 8051 microcontroller with aid of neat block diagram. **08**

   b. Draw the frame format of PSW register and explain each field. **06**

6. a. Discuss the following instructions with Example. **06**

   i. ADDC A, @Ri

   ii. SWAP A

   iii. RRC A

   b. Write an ALP to find larger number in single dimensional array of length N=5. **08**

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7. a. Explain the working of Timer-1 in Mode-1 configuration when GATE = 0 and GATE =1 with neat sketch. **06**

b. Discuss the steps to transmit data serial using TI flag. **06**

c. List interrupt available in 8051 with the interrupt address. **02**

8. a. Interface a DAC to 8051 microcontroller and develop C-program to generate Triangular waveform. **06**

b. Interface a 16x2 LCD in 8-bit mode to 8051 microcontroller. Use Port-2 for data and three line of Port-3 for control lines and develop C-program to display message "Welcome". **08**